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Bhuvann Vinodh Ram

Graduate AI / Machine Learning Engineer

Stamp 1G holder: eligible to work full-time in Ireland without a work permit

PROFILE

MSc AI for Medicine graduate (UCD) who builds and ships machine learning: CNNs, BiLSTMs, VAEs and gradient boosting on imaging, signal, text and genomics data, with SHAP explanations, leakage checks, tests and CI. Deploys models as Streamlit and Gradio apps and in-browser tools that cost nothing to run. Currently a Trust and Safety Associate at Accenture, Dublin.

EDUCATION

MSc Artificial Intelligence for Medicine and Medical Research — **2024 – 2025**
University College Dublin — GPA 3.37 / 4.2

BTech Computer Science and Medical Engineering — **2020 – 2024**
Sri Ramachandra Institute of Higher Education and Research, Chennai — CGPA 3.18 / 4

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, R (basic)
Machine learning	scikit-learn, XGBoost, ensembles, hyperparameter tuning, leakage checks, time-based validation
Deep learning	TensorFlow/Keras, PyTorch; CNNs, BiLSTM, VAEs, U-Net; transfer learning (ResNet50, VGG16, EfficientNet, MobileNet)
Vision and language	OpenCV, OpenCV.js, Detectron2, Tesseract.js; text classification, embeddings, TF-IDF retrieval
Explainability	SHAP, model interpretation
Shipping models	Streamlit, Gradio, GitHub Pages, Docker, GitHub Actions, pytest

EXPERIENCE

Trust and Safety Associate — **Mar 2026 – Present**
Accenture — Dublin, Ireland

- Evaluate, classify and moderate user-generated content against client platform policies to keep online communities safe and compliant.

Research Intern — **Apr 2025 – Sep 2025**
University College Dublin — Dublin, Ireland

- Designed and trained a variational autoencoder over RNA-seq, copy-number and methylation data, then clustered the latent space (K-Means, HDBSCAN).
- Explained the results with SHAP, differential expression and STRING networks, and assessed survival differences with Kaplan–Meier and log-rank tests.

Project Intern — **Jan 2024 – Jun 2024**
Centre for Toxicology and Development, SRIHER — Chennai, India

- Built a video-to-data pipeline for zebrafish behaviour analysis used in drug testing: tracked fish across tanks and layers, computed speed and velocity, and produced charts and reports.

Research Intern — **Dec 2022 – Feb 2023**
IIITDM Kancheepuram — Chennai, India

- Developed and compared CNN breast cancer classifiers using pre-trained VGG16, ResNet, MobileNet and EfficientNet models.

Intern Feb 2022 – Apr 2022
Smartail Pvt. Ltd. Chennai, India

- Containerised web applications with Docker and deployed them through GitLab CI/CD pipelines.
- Worked on image processing with Detectron2, including hyperparameter tuning.

PROJECTS

Tamil Thedal (தமிழ் தேடல்) 2026
JavaScript · OpenCV.js · Tesseract.js · Python ETL Live demo · GitHub

- Search engine for Tamil literature (1,549 verses) with sound-alike matching of romanised Tamil and a TF-IDF chat helper that runs fully in the browser.
- Inscription reader: OpenCV.js enhancement (CLAHE, adaptive threshold), letter segmentation and Hu-moment comparison with Tamil-Brahmi, Grantha and Tamil reference letters; Tesseract.js OCR for modern Tamil.

Conflict Risk Monitor 2026
Python · pandas · scikit-learn · GitHub Actions · JavaScript Live demo · GitHub

- Estimates the probability of 25+ battle deaths per country in the next three months; gradient boosting chosen on a validation period scored ROC AUC 0.989 and average precision 0.959 on unseen 2022–26 data.
- In-browser chat assistant (TF-IDF retrieval, no paid API) answers questions from the published data.

ECG heartbeat classification 2025
1D CNN · Keras GitHub

- 1D CNN classifying MIT-BIH heartbeats into five classes: 96.01% test accuracy on 21,892 held-out beats, reproduced by an automated GitHub Actions rerun.

Ovarian cancer survival prediction (HGSOC) 2025
XGBoost · SHAP · TCGA GitHub

- XGBoost on TCGA clinical and mutation data with SHAP explanations; found and removed follow-up variables that leaked the outcome, then reran the notebook end to end in GitHub Actions: test accuracy 0.78, ROC AUC 0.91.

Emotion classification from tweets 2025
BiLSTM · TensorFlow · Gradio GitHub

- Dual-input BiLSTM combining word embeddings with sentiment polarity: 90.63% test accuracy, served as a Gradio app.

Brain tumour classification from MRI 2025
CNN · OpenCV · Streamlit · pytest GitHub

- Four-class CNN (0.80 accuracy, 0.77 macro F1 on 2,258 test images) and ResNet50 transfer learning, deployed as a Streamlit app with unit tests and CI.

OTHER PROJECTS

- **Handwritten prescription reader** (Detectron2 · object detection, 2023): Detectron2 model that locates handwritten medicine names in prescription images, trained on a custom annotated dataset.
- **Multi-omics subtyping of colon cancer** (VAE · HDBSCAN · SHAP · survival analysis, 2025): Research project behind the UCD internship: VAE integration of three omics layers, clustering, and survival and pathway analysis, with negative findings reported honestly.
- **Teeth segmentation in dental X-rays** (U-Net, 2024): Semantic segmentation of panoramic dental X-rays with U-Net for diagnostic measurement.

AWARDS AND ACTIVITIES

- UCD Innovation Challenge 2024: Innovation Spark Award
- NASA Space Apps Challenge 2024: built a climate storytelling website with a team
- UCD Advantage Award
- First place, Tamil poetry competition (Agal Tamil Club)
- Second place, Palaestra intramural volleyball tournament